

The monomial Taylor tree in d dimensions

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Abstract

For a single monomial perturbation $\xi = a + c_0 u^\gamma$ each layer of the exponential tree (unit 62) is one tree term (unit 60) with shift γp and insertion order p , so the chart standard integral is the exact convergent series $\sum_{p \geq 0} \frac{(\beta c_0 c)^p}{p!} \int_{[0,b]^d} u^{h+\gamma p+kp} f(cu^k) du$ of box integrals with shifted exponents. This is the d -dimensional analogue of unit 26; by unit 60 every term has leading exponent in $\Lambda(h, k)$. Zero sorry/axiom.

1 Setting

The p -th layer integrand is $u^h(cu^k \cdot c_0 u^\gamma)^p f(cu^k) = c_0^p u^{h+\gamma p}(cu^k)^p f(cu^k)$; the only work is the algebra of products of powers over $\text{Fin } d$.

2 The things formalised

```
theorem treeLayer_monomial ( a b c c : ) {d : } (k h : Fin d → ) (p : ) :
  treeLayer a b c k h (fun u => c * i, u i ^ i) p
  = ( * c ) ^ p / p.factorial * treeTerm a b c k h (fun i => i * p) p
theorem boxIntegralXi_monomial_hasSum' ( a b c c : ) (h : 0 < ) (hb : 0 < b)
  (hc : 0 < c) {d : } (k h : Fin d → ) :
  HasSum (fun p => ( * c * c ) ^ p / p.factorial
    * boxIntegralFin a b c k (fun i => h i + i * p + k i * p))
  (boxIntegralXi a b c k h (fun u => c * i, u i ^ i))
```

3 Proof strategy

A pointwise identity of integrands (`pow_add`, `pow_mul`, `Finset.prod_mul_distrib`, `Finset.prod_pow`, `ring`) followed by pulling the constant c_0^p out of the integral; the series statements are `HasSum.congr_fun` of unit 62's theorem and unit 60's insertion identity.

4 Roadblocks and resolutions

- `pow_mul` rewrites $u^{p\gamma}$ as $(u^p)^\gamma$, which `Finset.prod_pow` cannot fold into $(\prod u^\gamma)^p$; writing the shift as $\gamma_i \cdot p$ makes the shapes agree.

5 What was Mathlib, what was new

Mathlib: `Finset.prod_pow`, `HasSum.congr_fun`. New: the monomial tree in d dimensions.

6 Lessons

Choosing the syntactic form of a natural-number exponent (γp versus $p\gamma$) to match the direction of the rewrite lemmas avoids a whole class of failed `simp` calls.

7 Outcome

The exponential tree, the tree terms and their exponents, and now their assembly for monomial perturbations are all formalised in d dimensions; the general polynomial case is a finite multinomial expansion of each layer into such tree terms.